HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 05 Name: _____ Marks: / 26

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. $\dfrac{2}{5-4x}-\dfrac{1}{5+4x}=$

A. $\dfrac{5+4x}{25-16x^2}$.

B. $\dfrac{5-4x}{25-16x^2}$.

C. $\dfrac{5+12x}{25-16x^2}$.

D. $\dfrac{5-12x}{25-16x^2}$.

(DSE-2023 PII Q02)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. $4^{n+2}3^{2n+4} =$

A. 6^{2n+4}.

B. 6^{4n+8}.

C. 12^{2n+4}.

D. 12^{3n+6}.

(DSE-2023 PII Q03)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. Let $g(x) = 13 - 5x^2$. If α is a constant, find $g(1 - 3\alpha)$.

A. $8 - 45\alpha^2$

B. $8 + 45\alpha^2$

C. $8 - 30\alpha + 45\alpha^2$

D. $8 + 30\alpha - 45\alpha^2$

(DSE-2023 PII Q08)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. Let $h(x) = ax^6 + 16x^3 + b$, where a and b are constants. If $h(x)$ is divisible by $2x - 3$, find the remainder when $h(x)$ is divided by $2x + 3$.

A. -108

B. -54

C. 54

D. 108

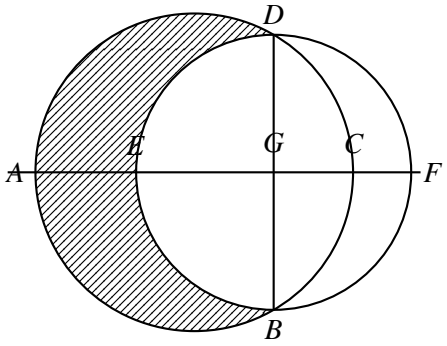
(DSE-2023 PII Q09)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

(DSE-2023 PII Q16)

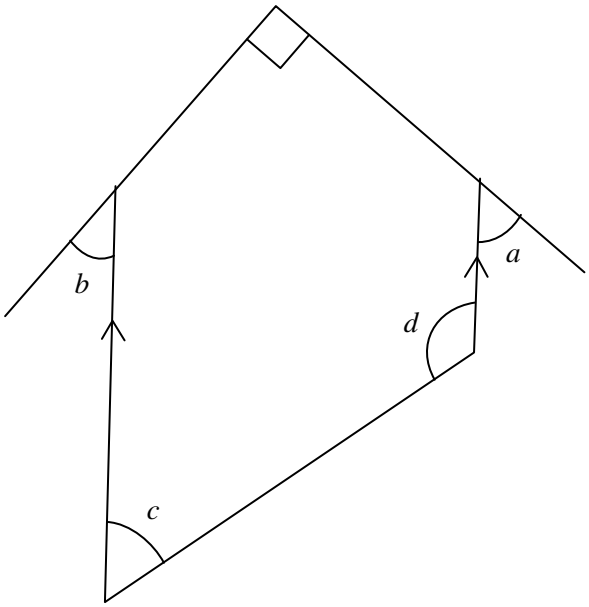
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. According to the figure, which of the following must be true?

?



I. $a+b=90^\circ$

II. $c+d=180^\circ$

III. $a+b+c=d$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. It is given that $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

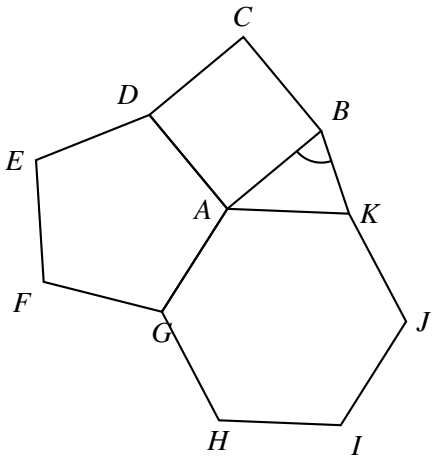
(DSE-2023 PII Q19)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. The figure shows the square $ABCD$, the regular pentagon $ADEFG$ and the regular hexagon $AGHIJK$. Find $\angle ABK$.



A. 69°

B. 72°

C. 74°

D. 75°

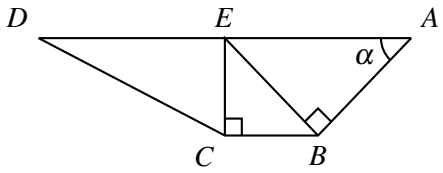
(DSE-2023 PII Q20)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.



A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

(DSE-2023 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.